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By substituting the values of $R_E = 6.4 \times 10^6\, m$ and g = 9.8 m s^{-2}, the orbital time period is obtained as T ≅ 85 minutes.

EXAMPLE 6.9

Moon is the natural satellite of Earth and it takes 27 days to go once around its orbit. Calculate the distance of the Moon from the surface of the Earth assuming the orbit of the Moon as circular.

Solution

We can use Kepler's third law,

$$T^2 = c\left(R_E + h\right)^3$$

$$T^{2/3} = c^{1/3}\left(R_E + h\right)$$

$$\left(\frac{T^2}{c}\right)^{1/3} = \left(R_E + h\right)$$

$$\left(\frac{T^2 GM_E}{4\pi^2}\right)^{\frac{1}{3}} = \left(R_E + h\right);$$

$$c = \frac{4\pi^2}{GM_E}$$

$$h = \left(\frac{T^2 GM_E}{4\pi^2}\right)^{1/3} - R_E$$

Here h is the distance of the Moon from the surface of the Earth. Here,

R_E – radius of the Earth = $6.4 \times 10^6\, m$

M_E – mass of the Earth = $6.02 \times 10^{24}\, kg$

G – Universal gravitational constant = $6.67 \times 10^{-11} \frac{Nm^2}{kg^2}$

By substituting these values, the distance to the Moon from the surface of the Earth is calculated to be $3.77 \times 10^5\, km$.

6.4.2 Energy of an Orbiting Satellite

The total energy of a satellite orbiting the Earth at a distance h from the surface of Earth is calculated as follows; The total energy of the satellite is the sum of its kinetic energy and the gravitational potential energy. The potential energy of the satellite is,

$$U = -\frac{GM_s M_E}{\left(R_E + h\right)} \qquad (6.63)$$

Here M_s - mass of the satellite, M_E - mass of the Earth, R_E - radius of the Earth.

The Kinetic energy of the satellite is

$$K.E = \frac{1}{2} M_s v^2 \qquad (6.64)$$

Here v is the orbital speed of the satellite and is equal to

$$v = \sqrt{\frac{GM_E}{\left(R_E + h\right)}} \qquad (6.65)$$

Substituting the value of v in (6.64), the kinetic energy of the satellite becomes,

$$K.E = \frac{1}{2}\frac{GM_E M_s}{\left(R_E + h\right)}$$